HYPERSPECTRAL OBJECT DETECTION CHALLENGE 2026

Ranked 6th out of 309 teams

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6th	309	Top 2%
final ranking	teams competing worldwide	of all participants

THE ACHIEVEMENT

Competing as a **one-person team**, I finished **6th of 309 teams** in Track 1 of the 2nd Hyperspectral Remote Sensing Data Processing and Application Challenge. The organisers have formally congratulated the result and **invited me to the ChineseWHISPERS Conference in Chengdu, China**, to be recognised alongside the other top teams.

WHAT THE COMPETITION WAS ABOUT

Ordinary cameras see three colours: red, green and blue. A **hyperspectral camera** sees many more; the one used here captures 16 narrow colour bands, so it can tell apart things that look identical to the human eye, such as a real apple and a plastic one.

The task was to build software that looks at these images and finds every object in them, drawing a box around each one and naming it. There were **18 kinds of object**, from fruit, eggs and shuttlecocks to people, cars and e-bikes. Many were real and fake look-alikes (real vs. plastic banana, real vs. wooden egg), and most were very small in the picture.

The challenge was hosted on Kaggle and organised by a group of universities including Nanjing University of Science and Technology, Shenzhen University, Wuhan University and the University of Cambridge.

HOW IT WAS JUDGED

- Every team trained on the same 3,000 labelled images.
- Final rankings used a **fresh set of 1,000 images released only in the last days**, with scores kept hidden until the end, so there was no way to tune for it.
- Strict fairness rules: **one single model only** (no combining several models), no extra hand-labelled data, and only 3 submissions a day.
- The top 6 teams had to hand over their full code and model so the organisers could check the result could be reproduced.

WHAT I BUILT

I adapted a modern object-detection model so it could read all 16 colour bands at once instead of the usual three, then trained and tuned it carefully to draw boxes that line up closely with small objects. The whole system (data preparation, training, and a script that reproduces the final submission exactly) was designed, built and run by me alone, on my own hardware.

WHY IT MATTERS

Hyperspectral imaging is used in farming, food-quality checks, environmental monitoring and remote sensing from satellites and drones. Placing in the top 2% of an international field of 309 teams shows the approach holds up on a real, hard problem.